Supplementary material: every table generated from the saved runs

Anonymous WACV Applications Track submission

Paper ID *****

All numbers are read from the run files by the code that builds this document. Cells are the mean over class partitions with the sample standard deviation as a subscript; brackets give a 95% t-interval over partitions.

Score at 30% of the parameters	RESISC45		EuroSAT-MS		PlantVillage		PlantDoc	
	P	PMP	P	PMP	P	PMP	P	PMP
Dense ResNet-18, S0 only	$87.2_{\pm 2.5}$		$64.7_{\pm 4.9}$		$91.4_{\pm 5.6}$		$55.4_{\pm 1.3}$	
Dense ResNet-18, S0+R+M+R	$87.0_{\pm 2.5}$		$64.2_{\pm 4.7}$		$91.3_{\pm 5.6}$		$55.5_{\pm 1.1}$	
Random	$80.1_{\pm 3.6}$	$81.1_{\pm 3.0}$	$57.8_{\pm 6.7}$	$58.4_{\pm 4.6}$	$85.6_{\pm 8.1}$	$86.8_{\pm 7.7}$	$46.6_{\pm 1.3}$	$46.7_{\pm 1.7}$
Filter ℓ_1	$81.0_{\pm 3.1}^{\uparrow}$	$81.5_{\pm 3.0}$	$57.9_{\pm 6.2}$	$59.4_{\pm 5.7}$	$85.8_{\pm 8.3}$	$86.4_{\pm 7.7}$	$48.7_{\pm 0.9}^{\uparrow}$	$48.1_{\pm 0.8}$
BN scale $\lvert\gamma\rvert$	–	$80.0_{\pm 3.1}^{\downarrow}$	–	$56.8_{\pm 4.9}$	$85.1_{\pm 8.9}$	$85.1_{\pm 8.7}$	$44.4_{\pm 1.7}^{\downarrow}$	$45.5_{\pm 1.0}$
Taylor	$82.5_{\pm 2.9}^{\uparrow}$	$82.4_{\pm 3.0}^{\uparrow}$	$60.2_{\pm 6.0}^{\uparrow}$	$59.7_{\pm 6.3}$	$88.3_{\pm 7.1}^{\uparrow}$	$87.1_{\pm 7.1}$	$47.9_{\pm 1.2}$	$48.5_{\pm 1.0}^{\uparrow}$
Gradient $\mathcal{G}$	–	$81.4_{\pm 3.3}$	–	$58.9_{\pm 5.6}$	$87.3_{\pm 7.2}$	$86.7_{\pm 7.9}$	$47.6_{\pm 1.5}$	$46.9_{\pm 2.5}$
Variance $\mathcal{V}$	–	$81.5_{\pm 3.0}$	–	$57.3_{\pm 4.4}$	$86.5_{\pm 6.9}$	$86.5_{\pm 7.3}$	$47.2_{\pm 1.4}$	$47.3_{\pm 1.3}$
Fisher $\mathcal{D}$	$79.6_{\pm 3.1}$	$79.9_{\pm 2.9}^{\downarrow}$	$58.1_{\pm 6.5}$	$58.4_{\pm 6.0}$	$85.2_{\pm 8.9}$	$85.3_{\pm 8.0}$	$46.8_{\pm 0.8}$	$46.4_{\pm 1.3}$
DACIS	$80.1_{\pm 3.0}$	$81.0_{\pm 3.3}$	$57.8_{\pm 5.3}$	$59.3_{\pm 4.9}$	$86.4_{\pm 7.7}$	$86.6_{\pm 7.5}$	$47.3_{\pm 1.4}$	$47.5_{\pm 1.6}$

Table 1. All eight scores at 30% of the parameters, schedules P and PMP.

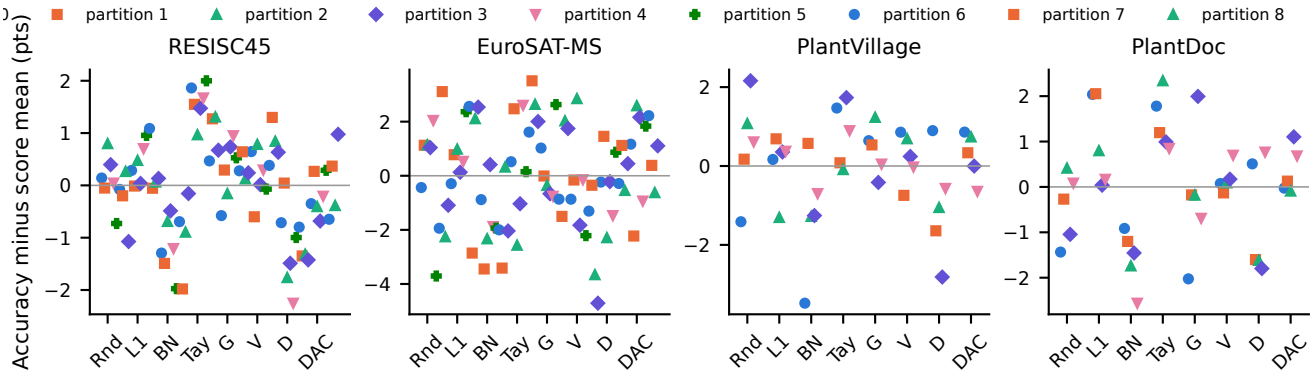


Figure 1. 5-shot accuracy of each score at 30% of the parameters (schedule PMP) on each class partition, relative to the mean over the eight scores; markers are partitions, offset horizontally for legibility. Rnd: random, L1: filter ℓ_1, BN: BatchNorm scale, Tay: Taylor, G, V, D: gradient, variance, Fisher, DAC: DACIS.

Schedule	Steps after S0
P	P(30%), R
P→M	P(30%), R, M
M→P	M, P(30%), R
P→G→P	P(60%), R, G, P(30%), R
PMP	P(60%), R, M, P(30%), R
PMPM	PMP, then M
P→M, 2×	P(30%), R, M with twice the episodes

Table 2. The seven schedules (P(x): prune to share x of the parameters; G: gradient probe).

Difference (a − b), points	RESISC45	EuroSAT-MS	PlantVillage	PlantDoc
Scores, schedule P				
DACIS − Random	−0.1 [−0.9,+0.8]	+0.1 [−2.6,+2.8]	+0.8 [−1.3,+3.0]	+0.7 [−1.2,+2.6]
DACIS − Filter ℓ_1	−0.9 [−1.5,−0.3]	−0.1 [−2.2,+2.1]	+0.6 [−1.9,+3.1]	−1.4 [−3.3,+0.5]
DACIS − Taylor	−2.4 [−3.0,−1.7]	−2.4 [−4.9,+0.1]	−1.9 [−2.9,−0.9]	−0.6 [−1.6,+0.4]
Scores, schedule PMP				
DACIS − Random	0.0 [−0.8,+0.7]	+0.9 [−1.3,+3.0]	−0.3 [−2.3,+1.8]	+0.8 [−0.4,+2.1]
DACIS − Filter ℓ_1	−0.4 [−1.0,+0.1]	−0.1 [−1.6,+1.4]	+0.2 [−1.3,+1.7]	−0.7 [−2.4,+1.1]
DACIS − Taylor	−1.4 [−1.9,−0.9]	−0.4 [−2.7,+1.9]	−0.6 [−1.9,+0.8]	−1.1 [−2.4,+0.3]
Episodic stage on the dense network				
Dense+M − Dense	−0.3 [−0.4,−0.1]	−0.8 [−1.5,0.0]	−0.3 [−0.6,+0.1]	0.0 [−0.4,+0.3]

Table 3. Paired differences between choices, in points.

Difference (a − b), points	RESISC45	EuroSAT-MS	PlantVillage	PlantDoc
Schedules, DACIS at 30%				
PMP − P	+0.9 [+0.4,+1.5]	+1.4 [−0.3,+3.2]	+0.1 [−0.7,+0.9]	+0.1 [−0.8,+1.1]
P→M − P	0.0 [−0.6,+0.5]	+0.7 [−0.9,+2.3]	−0.7 [−2.4,+1.0]	−0.8 [−1.6,0.0]
M→P − P	+0.9 [+0.3,+1.4]	+0.9 [−0.7,+2.4]	+0.3 [−1.2,+1.7]	−0.7 [−1.5,+0.2]
P→G→P (no weight update) − P	+1.2 [+0.6,+1.8]	+0.7 [−0.6,+2.0]	+0.3 [−0.4,+1.0]	−0.2 [−0.8,+0.5]
P→M→P→M − PMP	–	–	−0.1 [−1.2,+1.1]	+0.1 [−1.0,+1.2]
P→M (2× episodes) − P→M	–	–	−0.1 [−0.7,+0.4]	0.0 [−1.0,+0.9]

Table 4. Paired differences between schedules for DACIS at 30% of the parameters, in points.

	RESISC45			EuroSAT-MS			PlantVillage		
Model	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot
Dense ResNet-18	$71.1_{\pm 3.6}$	$87.2_{\pm 2.5}$	$89.5_{\pm 2.3}$	$51.3_{\pm 5.3}$	$64.7_{\pm 4.9}$	$67.6_{\pm 4.8}$	$79.5_{\pm 9.6}$	$91.4_{\pm 5.6}$	$93.1_{\pm 4.4}$
Random, PMP 30%	$64.6_{\pm 3.6}$	$81.1_{\pm 3.0}$	$84.0_{\pm 2.8}$	$46.6_{\pm 5.1}$	$58.4_{\pm 4.6}$	$61.4_{\pm 4.5}$	$73.6_{\pm 10.9}$	$86.8_{\pm 7.7}$	$88.9_{\pm 6.5}$
Filter ℓ_1, PMP 30%	$65.6_{\pm 3.5}$	$81.5_{\pm 3.0}$	$84.3_{\pm 2.8}$	$47.8_{\pm 6.2}$	$59.4_{\pm 5.7}$	$62.0_{\pm 5.5}$	$73.5_{\pm 11.1}$	$86.4_{\pm 7.7}$	$88.7_{\pm 6.4}$
Taylor, PMP 30%	$64.7_{\pm 3.8}$	$82.4_{\pm 3.0}$	$85.1_{\pm 2.8}$	$47.7_{\pm 6.5}$	$59.7_{\pm 6.3}$	$62.5_{\pm 6.0}$	$72.9_{\pm 11.0}$	$87.1_{\pm 7.1}$	$89.4_{\pm 5.8}$
DACIS, PMP 30%	$63.5_{\pm 3.9}$	$81.0_{\pm 3.3}$	$84.0_{\pm 2.9}$	$47.7_{\pm 5.8}$	$59.3_{\pm 4.9}$	$61.9_{\pm 4.7}$	$72.7_{\pm 10.8}$	$86.6_{\pm 7.5}$	$88.9_{\pm 6.3}$
MobileNetV3-Small	$73.1_{\pm 3.5}$	$88.1_{\pm 2.5}$	$90.3_{\pm 2.2}$	$60.2_{\pm 8.3}$	$74.7_{\pm 6.0}$	$78.0_{\pm 5.3}$	$78.0_{\pm 10.1}$	$91.1_{\pm 5.9}$	$93.1_{\pm 4.7}$

Table 5. 1-, 5- and 10-shot accuracy (%) of the main models on the three novel-class sets.

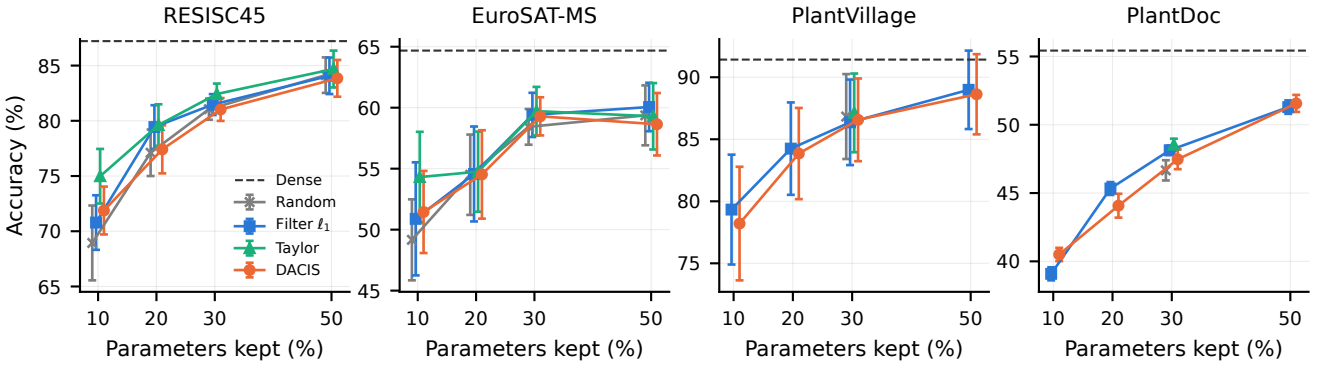


Figure 2. 5-shot accuracy against the share of parameters kept for four scores (schedule PMP); error bars are the standard error over partitions; dashed: dense network.

Schedule (DACIS, 30%), 5-shot	RESISC45	EuroSAT-MS	PlantVillage	PlantDoc
P	$80.1_{\pm 3.0}$	$57.8_{\pm 5.3}$	$86.4_{\pm 7.7}$	$47.3_{\pm 1.4}$
P→M	$80.0_{\pm 3.3}$	$58.5_{\pm 5.9}$	$85.8_{\pm 7.6}$	$46.5_{\pm 1.3}$
P→M (2×)	–	–	$85.6_{\pm 7.3}$	$46.4_{\pm 1.2}$
M→P	$80.9_{\pm 3.2}$	$58.7_{\pm 5.4}$	$86.7_{\pm 6.9}$	$46.7_{\pm 0.9}$
P→G→P	$81.3_{\pm 3.2}$	$58.6_{\pm 5.4}$	$86.8_{\pm 7.4}$	$47.1_{\pm 1.4}$
P→M→P (PMP)	$81.0_{\pm 3.3}$	$59.3_{\pm 4.9}$	$86.6_{\pm 7.5}$	$47.5_{\pm 1.6}$
P→M→P→M	–	–	$86.5_{\pm 7.4}$	$47.6_{\pm 1.6}$

Table 6. Schedules for DACIS at 30% of the parameters, 5-shot accuracy.

Params kept	Random	Filter ℓ_1	Taylor	DACIS
RESISC45				
50%	$84.1_{\pm 3.6}$	$84.1_{\pm 3.7}$	$84.7_{\pm 3.7}$	$83.8_{\pm 3.7}$
30%	$81.1_{\pm 3.0}$	$81.5_{\pm 3.0}$	$82.4_{\pm 3.0}$	$81.0_{\pm 3.3}$
20%	$77.1_{\pm 4.7}$	$79.4_{\pm 4.4}$	$79.6_{\pm 4.2}$	$77.4_{\pm 4.9}$
10%	$68.9_{\pm 7.6}$	$70.8_{\pm 5.5}$	$75.0_{\pm 5.5}$	$71.9_{\pm 4.8}$
EuroSAT-MS				
50%	$59.4_{\pm 5.5}$	$60.1_{\pm 4.4}$	$59.3_{\pm 6.1}$	$58.6_{\pm 5.7}$
30%	$58.4_{\pm 4.6}$	$59.4_{\pm 5.7}$	$59.7_{\pm 6.3}$	$59.3_{\pm 4.9}$
20%	$54.5_{\pm 7.4}$	$54.6_{\pm 8.7}$	$54.8_{\pm 7.3}$	$54.5_{\pm 8.1}$
10%	$49.2_{\pm 7.4}$	$50.9_{\pm 10.4}$	$54.3_{\pm 8.3}$	$51.4_{\pm 7.5}$
PlantVillage				
50%	–	$89.0_{\pm 7.1}$	–	$88.6_{\pm 7.2}$
30%	$86.8_{\pm 7.7}$	$86.4_{\pm 7.7}$	$87.1_{\pm 7.1}$	$86.6_{\pm 7.5}$
20%	–	$84.2_{\pm 8.3}$	–	$83.8_{\pm 8.2}$
10%	–	$79.3_{\pm 9.9}$	–	$78.2_{\pm 10.2}$
PlantDoc				
50%	–	$51.3_{\pm 1.1}$	–	$51.6_{\pm 1.4}$
30%	$46.7_{\pm 1.7}$	$48.1_{\pm 0.8}$	$48.5_{\pm 1.0}$	$47.5_{\pm 1.6}$
20%	–	$45.3_{\pm 1.0}$	–	$44.1_{\pm 2.0}$
10%	–	$39.1_{\pm 1.1}$	–	$40.5_{\pm 1.1}$

Table 7. 5-shot accuracy of four scores at four budgets (schedule PMP), one block per test setting.

PlantVillage 5-shot, schedule P, 30%	1 epoch	3 epochs	6 epochs
Random	$85.6_{\pm 8.1}$	$86.8_{\pm 8.1}$	$86.3_{\pm 8.2}$
Filter ℓ_1	$85.8_{\pm 8.3}$	$87.4_{\pm 7.3}$	$87.3_{\pm 7.2}$
Taylor	$88.3_{\pm 7.1}$	$88.2_{\pm 7.3}$	$87.8_{\pm 7.7}$
DACIS	$86.4_{\pm 7.7}$	$86.6_{\pm 7.7}$	$86.5_{\pm 7.7}$
Dense, no pruning		$91.4_{\pm 5.6}$	

Table 8. Recovery length after pruning (PlantVillage).

DACIS variant (PMP, 30%)	PV 5-shot	Δ vs. full
DACIS (full)	$86.6_{\pm 7.5}$	–
$\mathcal{G}$ only	$86.7_{\pm 7.9}$	+0.2 [−0.4,+0.7]
$\mathcal{V}$ only	$86.5_{\pm 7.3}$	−0.1 [−0.8,+0.7]
$\mathcal{D}$ only	$85.3_{\pm 8.0}$	−1.3 [−2.9,+0.3]
$\mathcal{G}$+$\mathcal{V}$	$86.9_{\pm 7.0}$	+0.4 [−0.7,+1.5]
$\mathcal{G}$+$\mathcal{D}$	$86.5_{\pm 7.5}$	−0.1 [−1.2,+1.1]
$\mathcal{V}$+$\mathcal{D}$	$86.2_{\pm 7.6}$	−0.4 [−1.5,+0.7]
no meta-gradient term	$85.6_{\pm 7.6}$	−0.9 [−1.7,−0.1]
same width at every depth	$86.1_{\pm 7.6}$	−0.5 [−1.2,+0.1]
min-max instead of ranks	$86.1_{\pm 7.9}$	−0.5 [−1.2,+0.2]

Table 9. DACIS components and design choices (PlantVillage).

$(\lambda_G, \lambda_V, \lambda_D)$	Val 1-shot	Val 5-shot	Novel 1-shot	Novel 5-shot
(0.3,0.2,0.5)	$74.3_{\pm 6.8}$	$88.5_{\pm 4.2}$	$73.3_{\pm 10.4}$	$86.9_{\pm 7.2}$
(0.33,0.33,0.33)	$73.7_{\pm 6.2}$	$88.1_{\pm 4.0}$	$73.2_{\pm 10.7}$	$86.9_{\pm 7.5}$
(0.5,0.2,0.3)	$73.6_{\pm 6.5}$	$87.9_{\pm 4.5}$	$72.3_{\pm 10.8}$	$86.2_{\pm 7.7}$
(0.2,0.5,0.3)	$74.3_{\pm 6.0}$	$88.5_{\pm 4.0}$	$72.7_{\pm 10.9}$	$86.5_{\pm 7.3}$
(0.2,0.2,0.6)	$74.3_{\pm 6.9}$	$88.9_{\pm 4.0}$	$72.4_{\pm 10.7}$	$86.2_{\pm 7.8}$
(0.1,0.1,0.8)	$74.3_{\pm 6.3}$	$88.2_{\pm 4.2}$	$72.3_{\pm 10.9}$	$86.1_{\pm 7.8}$
(0,0,1)	$74.1_{\pm 7.4}$	$88.0_{\pm 4.6}$	$71.8_{\pm 10.2}$	$85.9_{\pm 7.8}$
(0.5,0.5,0)	$74.1_{\pm 5.7}$	$88.8_{\pm 3.4}$	$73.0_{\pm 10.8}$	$86.9_{\pm 7.0}$

Table 10. Weights of DACIS on validation and novel classes (PlantVillage).

Episodic learning rate	PlantVillage val.		RESISC45 val.	
	1-shot	5-shot	1-shot	5-shot
none (S0 only)	86.7	96.1	74.2	90.1
0.0001	85.4	95.3	75.8	90.0
0.00025	84.9	95.0	76.1	89.8
0.0005	84.3	94.7	75.9	89.5
0.001	83.6	94.5	75.2	89.0
0.002	82.7	94.2	75.0	89.1
1e-05 (extra check)	–	–	74.5	90.1
2.5e-05 (extra check)	–	–	74.9	90.1
5e-05 (extra check)	–	–	75.3	90.1

Table 11. Learning rate of the episodic stage, 1- and 5-shot accuracy on the validation classes of the first partition. The PlantVillage column chose the value used everywhere; the RESISC45 sweep is a check.

EuroSAT-MS, novel classes	1-shot	5-shot	10-shot
13 bands, dense (S0)	$51.3_{\pm 5.3}$	$64.7_{\pm 4.9}$	$67.6_{\pm 4.8}$
RGB bands, dense (S0)	$53.7_{\pm 4.3}$	$68.7_{\pm 3.9}$	$71.4_{\pm 3.6}$
13 bands, dense (S0+R+M+R)	$50.8_{\pm 5.0}$	$64.2_{\pm 4.7}$	$67.0_{\pm 4.7}$
RGB bands, dense (S0+R+M+R)	$53.1_{\pm 4.5}$	$68.1_{\pm 3.8}$	$70.9_{\pm 3.4}$
13 bands, filter ℓ_1, PMP 30%	$47.8_{\pm 6.2}$	$59.4_{\pm 5.7}$	$62.0_{\pm 5.5}$
RGB bands, filter ℓ_1, PMP 30%	$49.8_{\pm 2.4}$	$63.9_{\pm 2.6}$	$66.6_{\pm 2.3}$
13 bands, DACIS, PMP 30%	$47.7_{\pm 5.8}$	$59.3_{\pm 4.9}$	$61.9_{\pm 4.7}$
RGB bands, DACIS, PMP 30%	$50.2_{\pm 3.5}$	$63.6_{\pm 3.9}$	$66.2_{\pm 3.7}$
13 bands, MobileNetV3-Small	$60.2_{\pm 8.3}$	$74.7_{\pm 6.0}$	$78.0_{\pm 5.3}$
RGB bands, MobileNetV3-Small	$58.3_{\pm 3.7}$	$74.7_{\pm 4.1}$	$78.0_{\pm 4.1}$
13 bands − RGB, dense (5-shot)		−6.6 [−17.4,+4.2]	
13 bands − RGB, DACIS PMP (5-shot)		−6.4 [−15.5,+2.6]	

Table 12. EuroSAT-MS with 13 bands against the three visible bands.

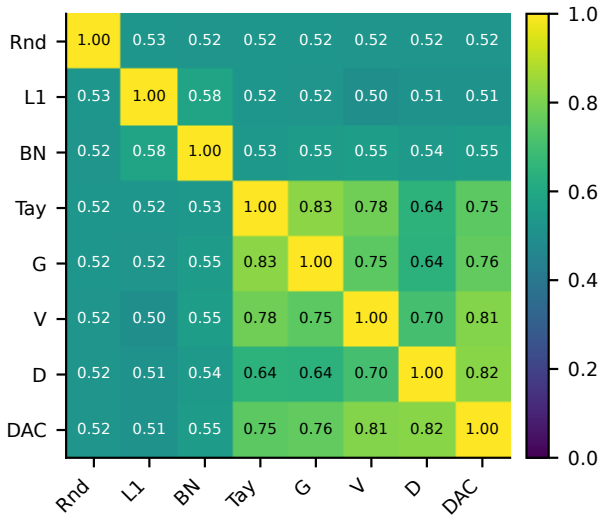


Figure 3. Overlap (Jaccard) of the channels kept by the first pruning step of different scores (RESISC45 if available, else PlantVillage).

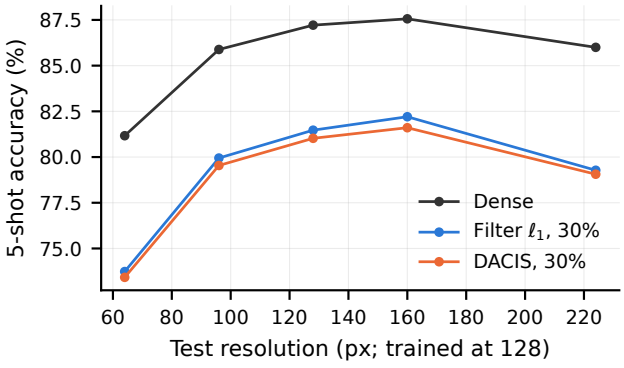


Figure 4. 5-shot accuracy when the test resolution differs from the training resolution (RESISC45 if available, else PlantVillage).